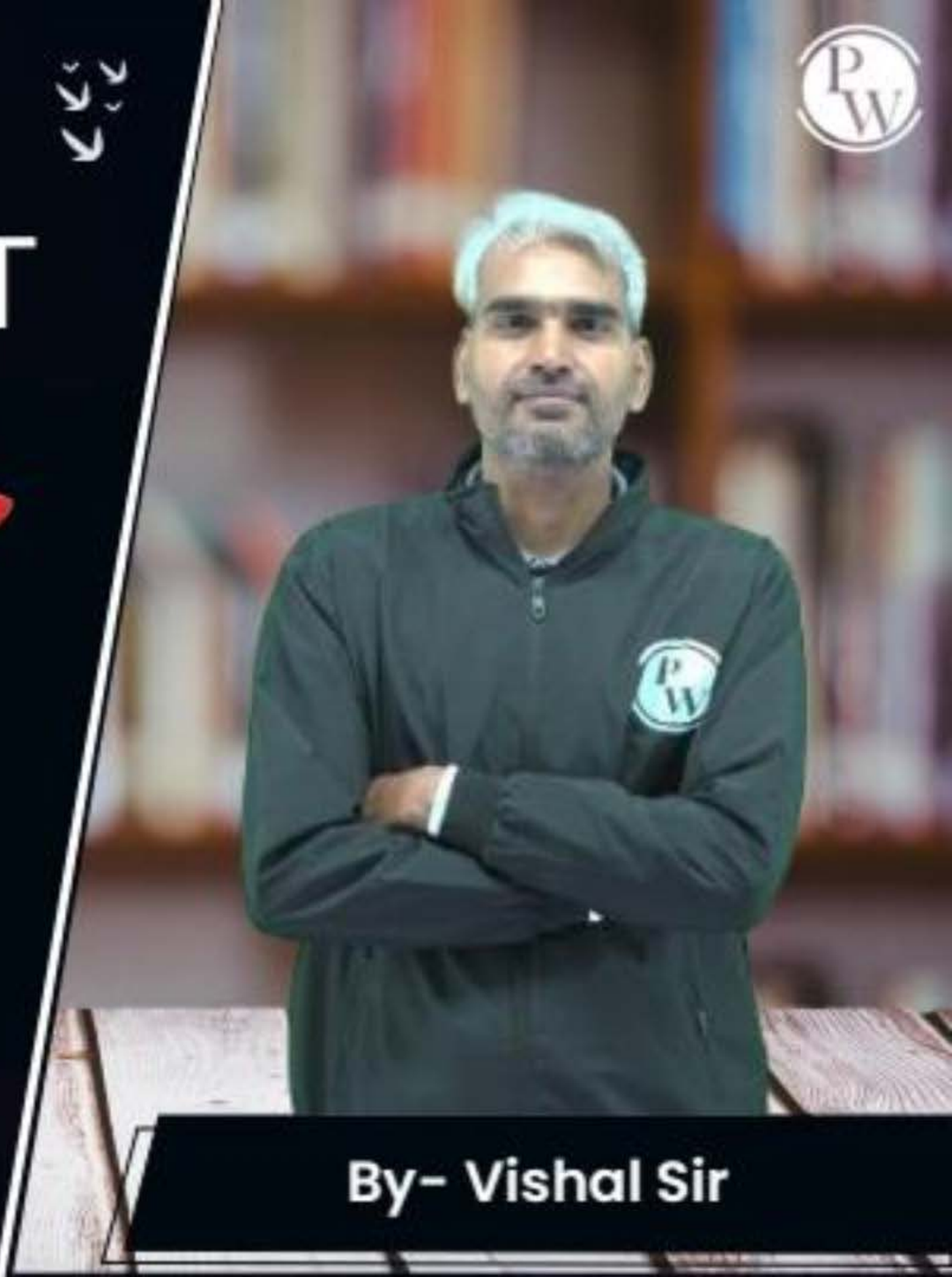
Computer Science & IT



Database Management System

Query Languages

Lecture No. 10



By- Vishal Sir

Recap of Previous Lecture



Topic Tuple relational calculus (TRC)

Slide

Topics to be Covered



Topic	Practice questions on Tuple relational calculus (TRC)

Slide



Topic : Unsafe TRC query

eg:

tuples Not belonging to the relation

$\{ t \mid t \notin \text{Relation} \}$

or

$\{ t \mid \neg(t \in \text{Relation}) \}$

Not the tuples belonging to relation

Unsafe TRC query may produce infinite number of tuples

Note: Power of Safe TRC query $\equiv$ Power of relational algebra

Power of TRC $\not\equiv$ Power of relational algebra

{Because of Unsafe TRC}

#Q. Which of the following relational calculus expressions is not safe?

True — it will be true only for the tuples belonging to relation R_1, ∴ query will produce finite no. of tuples

A. $\{t \mid \exists u \in R1\ (t[A] = u[A]) \wedge \neg \exists s \in R2\ (t[A] = s[A])\}$

Creation of attribute

B. $\{t \mid \forall u \in R1\ (u[A] = \text{“x”} \Rightarrow \exists s \in R2\ (t[A] = s[A] \wedge s[A] = u[A]))\}$

if $u.A = x$ then there must be a tuple S in rel^n R_2 (such that

C. $\{t \mid \neg(t \in R1)\}$

D. $\{t \mid \exists u \in R1\ (t[A] = u[A]) \wedge \exists s \in R2\ (t[A] = s[A])\}$

o/p

A
1
2

False

R_1

A	B
t.A = 1	
t.A = 2	
t.A = 3	
t.A	4

R_2

A	C
1	
2	

#Q. Consider the relation employee (name, sex, supervisorName) with name as the key. Supervisor Name gives the name of the supervisor of the employee under consideration. What does the following Tuple Relational Calculus query produce?

tuples from the Emp table will be selected only if it returns true

$$\{e.name \mid employee(e) \wedge (\forall x)\ [\neg employee(x) \vee x.supervisorName \neq e.name \vee x.sex = \text{“male”}]\}$$

Permanently false

this condn will always be true for the Employees those are not supervisors of any employee

True

it will be checked on when tuple $e \in$ Employee

{C, E}

A. Names of employees with a male supervisor.

B. Names of employees with no immediate male subordinates {A, C, D}

C. Names of employees with no immediate female subordinates {C, A, B, F}

D. Names of employees with a female supervisor. {A, B, F,}

Employee

Name	Sex	Sup. name
A	M	C
B	F	A
C	M	NULL
D	F	C
E	M	A
F	F	C

A, C, B,

#Q. Consider the relation employee (name, sex, supervisorName) with name as the key. Supervisor Name gives the name of the supervisor of the employee under consideration. What does the following Tuple Relational Calculus query produce?

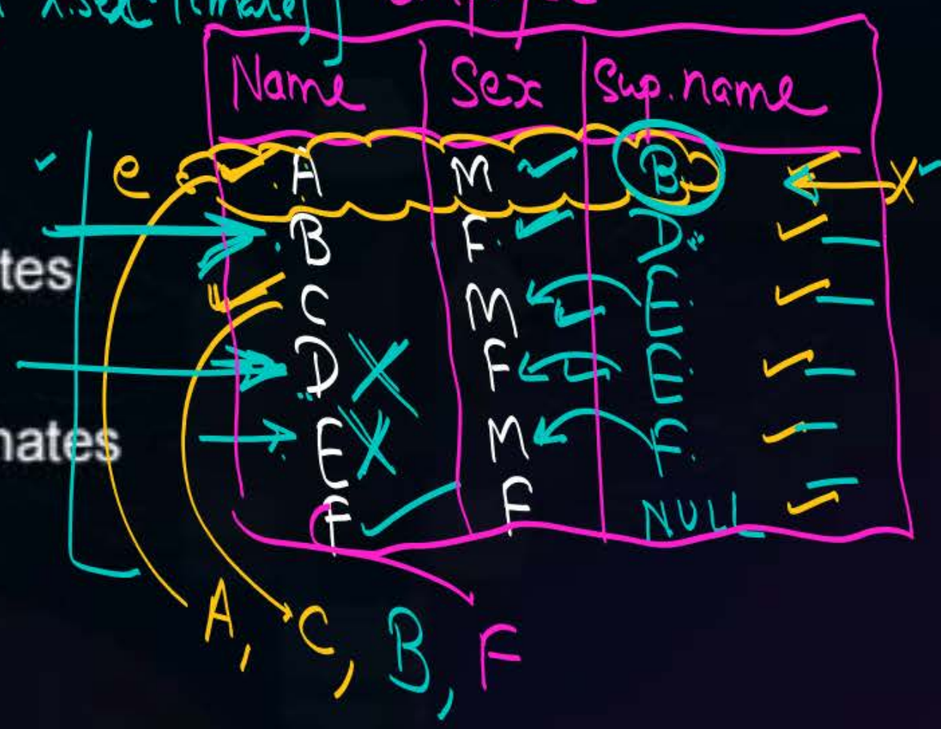
{e.name | employee(e)
∧ (∀x) [¬employee(x) ∨ x.supervisorName ≠ e.name ∨ x.sex = "male"]}.

A. Names of employees with a male supervisor.

B. Names of employees with no immediate male subordinates

C. Names of employees with no immediate female subordinates

D. Names of employees with a female supervisor.



#Q. Which of the following tuple relational calculus expression(s) is/are equivalent to $\forall t \in r(P(t))$?

I $\neg\exists t \in r(P(t))$

II $\exists t \notin r(P(t))$

III $\neg\exists t \in r(\neg P(t))$

IV $\neg\exists t \notin r(\neg P(t))$

for all tuple belonging to r Predicat 'P' must be true

Not for at least one tuple of r Predicate P should be false

A. I only

B. II only

C. III only

D. III and IV only

$\sim(\sim A) = A$

$\sim\left(\sim\left(\forall t \in r\left(P(t)\right)\right) \equiv \sim\left[\exists t \in r(\sim P(t)\right] \equiv \forall t \in r(P(t)\right)$

#Q. Consider The Following Relational Scheme

H.W.

Student (school-id, sch-roll-no, sname, saddress)

School (school-id, sch-name, sch-address, sch-phone)

Enrolment (school-id, sch-roll-no, erollno, examname)

ExamResult (Erollno, examname, marks)

Consider the following tuple relational calculus query

$$\{t \mid \exists E \in \text{Enrolment}\ t = E.\text{school-id} \wedge \{x \mid x \in \text{Enrolment} \wedge x.\text{school-id} = t \wedge (\exists B \in \text{ExamResult}\ B.\text{erollno} = x.\text{erollno} \wedge B.\text{examname} = x.\text{examname} \wedge B.\text{marks} > 35\} \div \{x \mid x \in \text{Enrolment} \wedge x.\text{school-id} = t\} * 100 > 35\}$$

If a student needs to score more than 35 marks to pass an exam what does the query return?

Options on the next Slide

#Q.

A. The empty set

B. schools with more than 35% of its students enrolled in some exam or the other

C. schools with a pass percentage above 35% over all exams taken together

D. schools with a pass percentage above 35% over each exam

#Q. H.W.

Consider a database that has the relation schemas
EMP(EmpId, EmpName, DeptId) and
DEPT(DeptName, DeptId).
Note that the DeptId can be permitted to a NULL in the relation EMP. Consider the following queries on the database expressed in tuple relational calculus.

I. $\{t \mid \exists u \in EMP\ (t[EMPName] = u[EmpName] \wedge \forall v \in DEPT\ (t[DeptId] \neq DeptId]))\}$

II. $\{t \mid \exists u \in EMP\ (t[EMPName] = u[EmpName] \wedge \exists v \in DEPT\ (t[DeptId] \neq DeptId]))\}$

III. $\{t \mid \exists u \in EMP\ (t[EMPName] = u[EmpName] \wedge \exists v \in DEPT\ (t[DeptId] = DeptId]))\}$

Which of the above queries are safe?

A. I and II only
B. I and III only
C. II and III only
D. I, II and III



2 mins Summary

Topic Tuple relational calculus

Slide

THANK - YOU

Slide